

```
DROP TABLE IF EXISTS bu_frd.model_controls_3m_2022_2025_v2;
```

```
CREATE TABLE bu_frd.model_controls_3m_2022_2025_v2
USING PARQUET
AS
```

```
WITH ranked_controls AS (
```

```
    SELECT
```

```
        c.*,
```

```
        ROW_NUMBER() OVER (
```

```
            PARTITION BY observation_date
```

```
            ORDER BY
```

```
                PMOD(
```

```
                    HASH(
```

```
                        CONCAT(
```

```
                            CAST(party_id AS STRING),
```

```
                            '|',
```

```
                            CAST(observation_date AS STRING),
```

```
                            '|CONTROL'
```

```
                        )
```

```
                    ),
```

```
                    2147483647
```

```
                ),
```

```
                party_id
```

```
            ) AS control_rank
```

```
FROM
```

```
    bu_frd.model_control_candidates_3m_2022_2025_v2 c
```

```
)
```

```
SELECT
```

```
    party_id,
```

```
    observation_date,
```

```
    observation_year,
```

```
    model_split,
```

```
    first_alert_date,
```

```
    0 AS label_3m,
```

```
    'CONTROL' AS cohort_role
```

```
FROM ranked_controls
```

```
WHERE control_rank <= positive_count * 20;
```

Validación 1 — ¿tenemos exactamente 20:1 por mes?

WITH positive_counts AS (

```
SELECT
  observation_date,
  COUNT(*) AS positives
```

```
FROM
  bu_frd.model_positive_anchor_3m_2022_2025_sample01_v2
```

```
GROUP BY observation_date
```

),

control_counts AS (

```
SELECT
  observation_date,
  COUNT(*) AS controls
```

```
FROM
  bu_frd.model_controls_3m_2022_2025_v2
```

```
GROUP BY observation_date
```

)

```
SELECT
  p.observation_date,

  p.positives,
  c.controls,

  ROUND(
    1.0 * c.controls / p.positives,
    2
  ) AS control_case_ratio
```

```
FROM positive_counts p
```

```
LEFT JOIN control_counts c
  ON p.observation_date = c.observation_date
```

```
ORDER BY p.observation_date;
```

Validación 2 — totales por split

```
SELECT
  model_split,
  COUNT(*) AS control_rows,
  COUNT(DISTINCT party_id) AS control_parties

FROM bu_frd.model_controls_3m_2022_2025_v2

GROUP BY model_split

ORDER BY model_split;
```

Validación 3 — ningún control repetido

```
SELECT
  COUNT(*) AS rows,
  COUNT(DISTINCT party_id) AS parties

FROM bu_frd.model_controls_3m_2022_2025_v2;
```

Validación 4 — ningún party puede ser positivo y control

```
SELECT
  COUNT(*) AS overlapping_parties

FROM
  bu_frd.model_controls_3m_2022_2025_v2 c

INNER JOIN
  bu_frd.model_positive_anchor_3m_2022_2025_sample01_v2 p

  ON c.party_id = p.party_id;
```

Validación 5 — comprobar nuevamente el label negativo

```
SELECT
  COUNT(*) AS invalid_controls

FROM
  bu_frd.model_controls_3m_2022_2025_v2

WHERE
  first_alert_date IS NOT NULL
```

```
AND first_alert_date <=
  LAST_DAY(
    ADD_MONTHS(observation_date,3)
  );
```

Paso 21E — crear una sola tabla de cohorte

```
DROP TABLE IF EXISTS bu_frd.model_cohort_3m_2022_2025_v2;
```

```
CREATE TABLE bu_frd.model_cohort_3m_2022_2025_v2
USING PARQUET
AS
```

```
SELECT
```

```
  party_id,
  observation_date,
  observation_year,
  model_split,
  first_alert_date,

  1 AS label_3m,
  'CASE' AS cohort_role
```

```
FROM
```

```
  bu_frd.model_positive_anchor_3m_2022_2025_sample01_v2
```

```
UNION ALL
```

```
SELECT
```

```
  party_id,
  observation_date,
  observation_year,
  model_split,
  first_alert_date,

  0 AS label_3m,
  'CONTROL' AS cohort_role
```

```
FROM
```

```
  bu_frd.model_controls_3m_2022_2025_v2;
```

Validación final del Paso 21

```
SELECT
```

```
  model_split,
```

cohort_role,
label_3m,

COUNT(*) AS rows,
COUNT(DISTINCT party_id) AS parties

FROM
bu_frd.model_cohort_3m_2022_2025_v2

GROUP BY
model_split,
cohort_role,
label_3m

ORDER BY
model_split,
label_3m DESC;

SELECT
COUNT(*) AS total_rows,
COUNT(DISTINCT party_id) AS total_parties,
SUM(label_3m) AS positives,
SUM(CASE WHEN label_3m = 0 THEN 1 ELSE 0 END) AS controls,

ROUND(
100.0 * SUM(label_3m) / COUNT(*),
4
) AS sampled_positive_rate_pct

FROM bu_frd.model_cohort_3m_2022_2025_v2;